Do Now · retrieval

Lesson 069 · Data Mining and Visualisation of Patterns

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is a backtracking branch?

Q2

What is an outlier?

Q3

What is the purpose of a scatter graph?

Q4

Array [2,3,4,5]: double elements at indexes 1 and 3. Give updated array and sum.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Define and predict — Data mining seeks patterns or associations across sets of data; define the question, prepare records and examine the results.

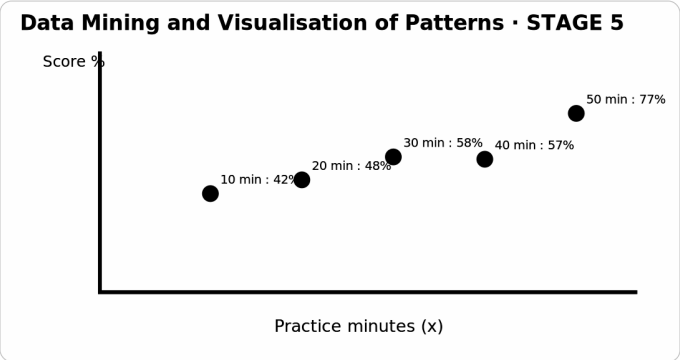
Core idea 2

Explain — Visualisation makes patterns and anomalies visible. Scatter charts suit pairs of numeric data; bar charts compare categories.

Core idea 3

Evaluate — A correlation does not establish cause. Sampling, confounders, missing values and measurement choices may mislead interpretation.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

data mining — Finding patterns in datasets.

scatter plot — Pairs of numeric variables plotted as points.

outlier — Unusually distant observation.

correlation — Association, not proof of causation.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

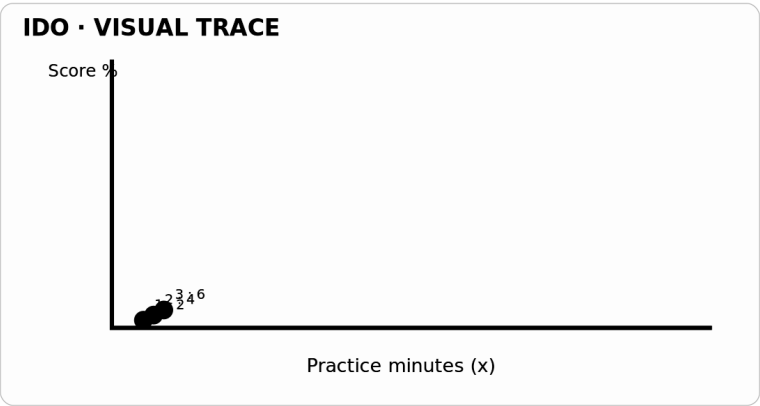
Problem

For points (1,2),(2,4),(3,6), state trend and one limitation.

Follow the worked method

1. Follow the reasoning — Separate observation from causal explanation.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Follow the reasoning
The value96 pulls mean away from three clustered lower scores.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

For points (1,2),(2,4),(3,6), choose a suitable chart and describe association. [4]

Question 2

For [40,42,46,96] calculate mean and median with working. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Evaluate mining a school's student-usage logs to predict attainment. Address fairness, data quality and privacy. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A rising scatter trend proves that extra practice alone caused higher marks.

Use key vocabulary from memory

1. Explain data mining in your own words.

2. Explain scatter plot in your own words.

3. Explain outlier in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.